KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · MATH

The Bridge Design Studio

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Shapes in Bridges

Ximena is spending the week at a bridge design studio in Pittsburgh, Pennsylvania. Warm up before the lesson, and show your work. Some problems may feel new, and that is okay!

1. Review (Week 23): a line plot of bolt lengths shows 3 X's at 3/8 inch and 2 X's at 3/4 inch. a) What is the total length of all 5 bolts? b) How much longer is the longest bolt than the shortest?

2. Draw a triangle that has one right angle. Mark the right angle with a small square.

3. How many sides and how many angles does a quadrilateral have? Name two quadrilaterals you already know.

4. Which kind of lines never cross, no matter how long they get?

   parallel lines | perpendicular lines | intersecting lines

5. Draw a shape that has exactly one line of symmetry. Draw the line of symmetry as a dashed line.

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The Bridge Design Studio

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Checking the Design

Name every triangle two ways, use the family tree, and show your work.

1. Classify each triangle by its sides AND by its angles. a) sides of 6, 6, and 6 feet b) angles of 90°, 30°, and 60°, with no equal sides c) angles of 110°, 35°, and 35°, with two equal sides

2. Two steps: an isosceles triangle brace has a perimeter of 31 feet. Its third side, the one that is not equal to the others, is 7 feet. How long is each equal side?

3. How many lines of symmetry? a) a square b) a rectangle that is not a square c) an equilateral triangle d) a scalene triangle Write your answers, and draw the lines for a) and b).

4. Find the mistake: Xander says every rectangle is a square, because both shapes have four right angles. In two or three complete sentences, explain his mistake.

★ **Challenge:** Write always, sometimes, or never. a) A rectangle is a square. b) A rhombus is a parallelogram. c) A triangle has a pair of parallel sides. Then draw a right trapezoid and mark its parallel sides with arrows.

Looking Ahead: Week 25: Unit 5 Checkpoint. Review the coordinate plane and number patterns, measurement conversions, line plots with fractions, and classifying two-dimensional figures.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · READING

Nothing but Fish and Fog

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Reading Between the Lines

Today you will read a story set in Sitka, Alaska, and a letter written by its main character. Show what you already know. It is fine to guess!

1. Review (Week 23): Name the kind of figurative language and tell what it really means: "The wind howled at the windows all night."

2. Read: "Arden stomped into the kitchen, dropped his muddy boots by the door, and would not look at anyone." How does Arden feel? Which clues tell you?

3. Which word is a **character trait** (how someone usually is), not a feeling that passes quickly?

   Sleepy | Generous | Surprised

4. Think of a character from a book or movie who changes. What were they like at the start, and what were they like at the end?

5. What is one thing in your own town that you think is boring? Write one sentence about what a visitor might notice about it instead.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · READING

Nothing but Fish and Fog

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Character Change Detective

Write in complete sentences with correct capitals, commas, and quotation marks.

1. Read: "When Grandpa handed Anya a knife to clean her first fish, she wrinkled her nose. Then she rolled up her sleeves." Make an inference about Anya, and give the clue.

2. In sentences: what is Anya like at the beginning (one trait, one clue)? What changes her? What is she like at the end (one trait, one clue)?

3. Find the mistake. Anders wrote: "Grandpa is angry at Anya, because he will not help her reel in the fish." Why is this inference wrong? Use evidence.

4. Be the editor. Rewrite each sentence correctly. a) However Anya forgot her phone. b) Arvid asked, "can you show me the whale?"

★ **Challenge:** In 3 or 4 sentences, show (do not tell) how Arvid changes on his first day on Grandpa's boat.

Looking Ahead: Week 25: Unit 5 Checkpoint. Review text structures, firsthand and secondhand accounts, figurative language, and character change and inference.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · SCIENCE

Earth's Moving Plates: Earthquakes, Volcanoes and Mountains

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 23):** Name the two main groups of decomposers. Then explain where the matter in a dead leaf goes when they break it down.

2. What do you think causes an earthquake? Write your best idea.

3. Name any volcano or mountain range you have heard of. Where is it?

4. Draw what you think is under the ground beneath a volcano. Label your drawing.

5. Predict: Do you think the continents have always been in the same places? Why or why not?

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · SCIENCE

Earth's Moving Plates: Earthquakes, Volcanoes and Mountains

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Plates in Motion

Use evidence from today's lesson. Write in complete sentences.

1. **Name the boundary** (divergent, convergent or transform): (a) the Andes rise as an ocean plate sinks under South America; (b) Iceland slowly splits; (c) land slides past land along the San Andreas fault.

2. **Read the data.** Alaska 1964 was magnitude 9.2; Mount St. Helens 1980 was magnitude 5.1. What is the difference? Which released more energy, and how do you know?

3. **Two-step problem.** The San Andreas fault moves about 5 centimeters a year. How many centimeters in 30 years? How many meters is that?

4. **Engineering.** Explain one way engineers help buildings survive earthquakes and one way ShakeAlert helps people.

★ **Challenge: Find the mistake.** "Hawaii has volcanoes because it is on a plate boundary." Explain what is wrong, then sketch where Hawaii's volcanoes form.

Looking Ahead: Next week is the Unit 5 Science Checkpoint, reviewing Weeks 21 to 24: chemical changes, senses and signals, decomposers and moving plates.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · SOCIAL STUDIES

World War II at Home and Abroad (1941–1945)

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Write what you know right now. Guesses are fine; this is your starting point.

1. **Review (Week 23):** Name one New Deal program (for example, the Civilian Conservation Corps or Social Security). Who did it help, and who did it leave out?

2. What do you already know about World War II? Write two facts or guesses.

3. Prediction: When millions of men left home to fight, who do you think did their jobs in factories and shipyards?

4. Why might a country treat some of its own people unfairly during a war? Explain your thinking.

5. Write one question you have about World War II.

KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 24 · SOCIAL STUDIES

World War II at Home and Abroad (1941–1945)

Name: _____ **Date:** _____

POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Freedom at Home and Abroad

Use evidence from today's lesson. Write in complete sentences.

1. **Put in order with years:** Civil Liberties Act apology; Germany invades Poland; Executive Order 9066; attack on Pearl Harbor.

2. **Find the mistake.** "Every American was treated as an equal during World War II." Rewrite it so it is true, using two examples from the lesson.

3. **Compare two sources.** The 1942 exclusion order said "both alien and non-alien." The 1988 Civil Liberties Act said Congress "apologizes on behalf of the Nation." How did the government's view change?

4. Choose Doris Miller, Chester Nez, Benjamin O. Davis Jr., Fred Korematsu or Aki Kurose. What unfairness did this person face, and how did he or she respond?

★ **Challenge:** Write a short paragraph (3 or 4 sentences) explaining what the "Double V" campaign meant and why it mattered. Check your capital letters, commas and end punctuation.

Looking Ahead: Next week is the Unit 5 Social Studies Checkpoint on Weeks 21 to 24. Bring this question: during war and hard times, who got help and freedom, and who had to fight for it?